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Occurrence and antimicrobial resistance of *Campylobacter* isolates from broiler chicken and slaughter house environment in India

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ABSTRACT

Campylobacteriosis is among the most frequently reported foodborne zoonoses. A total of 848 samples were screened for *Campylobacter* spp. and occurrence was found to be 8.7%, 2.3% and 1.65% in broiler cecum samples, chicken meat samples and slaughter house environment swabs, respectively. High level of antimicrobial resistance was found against tetracycline (64.1%), doxycycline (54.4%), ampicillin (46.6%), nalidixic acid (42.7%), kanamycin (35.9%), and ciprofloxacin (33.33%). Resistance to co-amoxiclav (19.4%) and erythromycin (21.4%) was less common. The MAR index of the isolates was in the range of 0.11–0.78. Multi-drug resistance was observed in 54.4% of the isolates, with 53.2% *C. jejuni* and 55.3% *C. coli* isolates found resistant against three or more classes of antimicrobials. Presence of mutations in *gyrA* and 23S rRNA genes was investigated, which revealed that all the fluoroquinolone resistant isolates possessed Thr-86-Ile point mutation, whereas only 68% of erythromycin resistant isolates had A2075G mutation. The *tetO* gene was present in 91.7% tetracycline resistant isolates and $bla_{\text{OXA-61}}$ gene was detected in 97.9% of the ampicillin resistant isolates. The occurrence of antimicrobial resistant *Campylobacter* spp. in broiler chicken samples and slaughter house settings is a public health risk and calls for judicial use of antimicrobials.

KEYWORDS
Campylobacter; broiler; antimicrobial resistance; resistance mechanisms

Introduction

Campylobacteriosis is among the most frequently reported zoonotic infections worldwide. *Campylobacter* spp. have been attributed globally as one of the four key causes of diarrheal diseases.[1] Campylobacters are present in the intestinal tracts of a vast majority of birds and animals used in food production, among which poultry is considered as a major source to humans. The horizontal spread of *Campylobacter* spp. occurs in the flock mainly by fecal-oral route and the infected birds continue to harbor the organism until slaughter.[2] During slaughter the intestinal contents may spill over onto raw finished products and the slaughter house environment, thus enabling their transmission and survival throughout the supply chains.[3]

Campylobacterioses in man result in an acute gastrointestinal illness characterized by diarrhea, fever and abdominal cramps. Post-infection manifestations such as Guillain-Barré syndrome, an acute demyelinating disease characterized by ascending paralysis, are known to occur in 1 in 1000 patients exposed to *Campylobacter* infection.[4]

Campylobacter spp. infections are self-limiting in majority of the cases and in cases where treatment is essential, fluoroquinolones and macrolides are the drugs of choice.[5] The development and spread of antimicrobial resistance in zoonotic bacteria such as *Campylobacter* that are exposed to antimicrobials in both animal production and human medicine has become a global public health problem.[6] Resistance to azithromycin has remained stable in recent years in the US with 3% of *C. jejuni* and 7% of *C. coli* isolates reported as resistant, while 8% of *C. jejuni* and 38% of *C. coli* isolates were resistant to ciprofloxacin.[7] Sporadic studies conducted in different parts of India have revealed the presence of *Campylobacter* spp. in various species of animals and birds, foods and raw milk samples.[8–10] Poultry farming is one of the fastest growing segments of the agricultural sector in India and the growth is driven by rising per capita income of the middle class.[11]

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The mechanisms of drug resistance in *Campylobacter* spp. have been characterized for nearly all classes of antimicrobials. Single point mutations in the quinolone resistance determining region (QRDR) of *gyr*A gene[12,13] and in the domain V of the 23S rRNA gene[14,15] are responsible for conferring resistance against fluoroquinolones and macrolides, respectively. Resistance to tetracycline is primarily mediated by the plasmid-encoded *tetO* gene. Drug inactivation has been attributed as a common mechanism for resistance to other antimicrobials such as aminoglycosides, β-lactams, and chloramphenicol.[12,16] The efflux pump *cme*ABC is associated with resistance to multiple drugs such as macrolides, fluoroquinolones, and tetracycline.[17] Multidrug resistance (MDR) in *Campylobacter* spp. has shown an increasing trend worldwide, especially within the food chain.[3,8,18,19] Monitoring of implications of antimicrobial resistance is highly relevant to public health and international trade.

The increasing trend of antimicrobial resistance has prompted the use of antimicrobial susceptibility testing in routine clinical practice. The slow growth and fastidious requirement of *Campylobacter* cause difficulties in routine antimicrobial susceptibility testing. Several methods including disk diffusion, Etest®, broth and agar dilution methods are available for *Campylobacter* spp. and the agar dilution methods are currently acceptable.[20] Despite its public health importance, surveillance of antimicrobial resistance for *Campylobacter* is inadequate in India. The identification of antimicrobial resistant strains, their distribution and the associated risk factors could provide important information for policy decisions regarding surveillance and judicious use of antimicrobials in both humans and agriculture. This study aims to determine the prevalence of *Campylobacter* spp. in retail broiler markets and slaughter house environment and to determine the antimicrobial resistance pattern at phenotypic and genotypic levels.

Materials and methods

Collection of samples

A total of 848 samples including poultry cecum ($n = 441$), chicken meat ($n = 275$) and swabs/sewage samples from slaughter house environment ($n = 132$) were collected randomly from retail meat shops and poultry slaughter houses in Bareilly, Junagadh and Bengaluru regions, India. Raw chicken meat samples (50 g each) and whole cecum were collected aseptically in sampling bags (Whirl-Pak®, Nesco). The swabs from chopping boards, knives and weighing scale were collected using sterile cotton swabs and transferred to 25 ml Bolton broth. Approx. 50 ml of sewage sample from the slaughter house was collected in sterile centrifuge tubes. The samples were dispatched to the laboratory under cold chain and processed within 4 h of collection to ensure that the organisms remain viable and culturable.

Isolation and identification of *Campylobacter* spp.

The isolation and identification of the *Campylobacter* spp. was carried out as per ISO 10272-1:2017. The chicken meat (10 g each) samples were homogenized in buffered peptone water and 1 ml of the homogenate was inoculated into 9 ml Bolton broth and incubated at 37 °C for 4 h (pre-enrichment) followed by 42 °C for 40 h (enrichment). The sewage samples were centrifuged for 5 min at 5000 rpm, supernatant discarded and the pellet was suspended in 9 ml Bolton broth. The swab samples were enriched in 9 ml Bolton broth as per the conditions mentioned above. Post enrichment, 10 µl of the enriched broth was plated on modified charcoal cefaperazone deoxycholate agar (mCCDA) (HiMedia, India) and incubated at 42 °C for 48 h under microaerobic conditions (CampyPak, BBL, USA). The cecal samples were cut open to scrape the inner walls using a loop and the contents were streaked directly onto a mCCDA plate and incubated under microaerobic conditions at 42 °C for 48 h. Isolates showing typical growth pattern were subjected to Gram's staining, motility test, aerobic growth test and standard biochemical tests.[21,22] Isolates that were Gram negative, showed cork-screw morphology, did not show growth in aerobic conditions at 42 °C or under microaerobic conditions at 25 °C, and were positive for oxidase, catalase and indoxyl acetate hydrolysis tests were considered as *Campylobacter* spp.

Molecular confirmation of the isolates

The genomic DNA of presumptively positive isolates was extracted from a single colony using DNeasy Blood and Tissue Kit (QIAGEN, USA). The isolates were identified at the genus level by PCR assay targeting 16S rRNA gene[23] and at species level using a multiplex PCR assay targeting the lipid gene *lpx*A.[24] The reaction mixture (25 µl) comprised 2.5 µl of 10X Dream *Taq* Buffer, 2.5 µl of 2 mM of each dNTP, 15 pmol of each primer, 1 U *Taq* Polymerase (Thermo

Fisher, USA), 3 µl of bacterial DNA and nuclease-free water to make up to 25 µl, and reactions were carried out in a thermalcycler (Eppendorf, Germany). The reference strain *C. jejuni* (ATCC 33560) was used as a positive control. The PCR products were analyzed by electrophoresis on 1.5% agarose gel (Amresco, USA) and the gel pictures were documented using Gel Doc IT imaging System (UVP, UK). All confirmed isolates were preserved in Brain Heart Infusion (BHI) Broth with 20% glycerol and stored at −20 °C until further use.

Antimicrobial susceptibility testing

Antimicrobial susceptibility was performed by broth microdilution assay against eight antibiotics *viz*., Ampicillin (AMP), co-amoxiclav (COA), ciprofloxacin (CIP), doxycycline (DOX), erythromycin (ERY), kanamycin (KAN), nalidixic acid (NA) and tetracycline (TET).

The broth microdilution assay was carried out in a 96 well cell culture plate with double diluted antibiotics. Cation-adjusted Mueller–Hinton broth (CAMHB) with 2.5% sheep blood was used as the growth medium and an inoculum of 0.5 McFarland turbidity was used for testing. The column 11 (CAMHB with 2.5% sheep blood and bacterial inoculum) and column 12 (MHB with 2.5% sheep blood) of the 96 well plate acted as the growth control and sterility control, respectively to assure suitability of culture and aseptic condition of the media. The plates were incubated at 37 °C for 48 h under microaerobic environment and results read out and expressed in MIC (µg/ml). The antimicrobial resistance was interpreted as recommended by the NARMS report[25] for *Campylobacter* spp. and CLSI (2017) guidelines. *C. jejuni* (ATCC 33560) was used for quality control at every batch of testing. The isolates were considered as multi-drug resistant (MDR) when found resistant against three or more classes of antimicrobials. The multiple antibiotic resistance index (MAR) of each isolate was calculated using the equation MAR index = a/b, where ‘a’ is the total number of antibiotics to which the isolate was resistant and ‘b’ refers to total number of antibiotics tested.[26]

Detection of genes associated with antimicrobial resistance

The genes associated with antimicrobial resistance against four classes of antibiotics *viz.,* fluoroquinolones (QRDR of *gyr*A), macrolides (23S rRNA), tetracyclines (*tet*O) and beta-lactams ($bla_{\text{OXA-61}}$) were detected by PCR. A fragment of the quinolone resistance determining region (QRDR) of *gyr*A was amplified as described previously[13] to generate a 673 bp product. A panel of 22 isolates, which were resistant against both nalidixic acid and ciprofloxacin, along with 10 susceptible isolates were sequenced commercially (Eurofins, India). The sequences were aligned (BioEdit Sequence Alignment Editor, Hall, 1999) and compared with previously reported sequences for presence of amino acid substitution at codon 86.

The 23S rRNA gene was amplified[14] to obtain a 714 bp product. The products of 32 isolates (19 resistant and 13 susceptible to Erythromycin) were sequenced as above and compared with previously reported sequences for nucleotide substitution.

The *tetO*[27] and $bla_{\text{OXA-61}}$[28] genes were amplified as previously described to generate amplicons of 559 bp and 281 bp, respectively. All the PCR products were analyzed by electrophoresis on 1.5% agarose gel (Amresco, USA) at 90 V for 45 min, with 1x TAE as running buffer. The loading mixture consisted of 8 µl of the amplified product mixed with 2 µl of 6X loading dye (Thermo Fisher, USA). DNA bands were stained with ethidium bromide (0.05 µg/ml), visualized and documented using Gel Doc IT imaging System (UVP, UK).

Results

Occurrence of *Campylobacter* spp.

Out of the 848 samples screened, 12.74% ($n = 108$) were found presumptively positive for *Campylobacter* spp. by isolation and biochemical assays. The isolates were confirmed by PCR and identified as *C. jejuni* ($n = 47$) and *C. coli* ($n = 56$), with 5 isolates positive for both the species and represented mixed cultures (Table 1). The occurrence was highest among chicken cecum samples (8.7%, $n = 74$), followed by chicken

Table 1. Occurrence of *Campylobacter* spp. in different sources.

	Percent (*n*/total samples)			
Source of samples	Total positive	*C. jejuni*	*C. coli*	*C. jejuni* + *C. coli*
Poultry cecum	8.7% (74/848)	3.8% (32/848)	4.5% (38/848)	0.5% (4/848)
Chicken meat	2.4% (20/848)	1.1% (9/848)	1.2% (10/848)	0.1% (1/848)
Slaughter house environment swabs/sewage	1.7% (14/848)	0.7% (6/848)	0.9% (8/848)	–
Total	12.7% (108/848)	5.5% (47/848)	6.6% (56/848)	0.6% (5/848)

Table 2. Species-wise pattern of antimicrobial resistance.

		C. jejuni			*C. coli*		
CLSI Class	Antimicrobial agent	Resistance % (*n/N*)[a]	MIC_{50} (μg/ml)	MIC_{90} (μg/ml)	Resistance % (*n/N*)[a]	MIC_{50} (μg/ml)	MIC_{90} (μg/ml)
Penicillins	Ampicillin	59.6 (28/47)	16	64	35.7 (20/56)	2	64
	Co-amoxiclav	19.1 (9/47)	2	32	19.6 (11 /56)	2	32
Fluoroquinolones	Ciprofloxacin	31.9 (15/47)	0.5	32	35.7 (20/56)	1	32
	Nalidixic acid	36.2 (17/47)	32	64	48.2 (27/56)	8	64
Macrolide	Erythromycin	27.7 (13/47)	2	32	16.1 (9/56)	2	128
Aminoglycoside	Kanamycin	38.3 (18/47)	2	32	33.9 (19/56)	2	64
Tetracycline	Doxycycline	46.8 (22/47)	8	32	60.7 (34/56)	8	32
	Tetracycline	68.1 (32/47)	32	128	60.7 (34/56)	32	128

[a]*N* = Total no. of isolates; *n* = no. of resistant isolates.

Table 3. Source-wise pattern of antimicrobial resistance.

		Resistance % (*n/N*)[a]		
CLSI Class	Antimicrobial agent	Chicken meat	Poultry cecum	Slaughter house environment swabs/sewage
Penicillins	Ampicillin	57.9 (11/19)	37.1 (26/70)	78.6 (11/14)
	Co-amoxiclav	31.6 (6/19)	11.4 (8/70)	42.9 (6/14)
Fluoroquinolones	Ciprofloxacin	31.6 (6/19)	32.9 (23/70)	42.9 (6/14)
	Nalidixic acid	52.6 (10/19)	40.0 (28/70)	42.9 (6/14)
Macrolide	Erythromycin	26.3 (5/19)	24.3 (17/70)	0 (0/14)
Aminoglycoside	Kanamycin	36.8 (7/19)	35.7 (25/70)	35.7 (5/14)
Tetracycline	Doxycycline	63.2 (12/19)	48.6 (34/70)	71.4 (10/14)
	Tetracycline	68.4 (13/19)	61.4 (43/70)	71.4 (10/14)

[a]*N* = Total no. of isolates; *n* = no. of resistant isolates.

meat (2.3%, $n = 20$) and slaughter house environment samples (1.65%, $n = 14$). The five mixed cultures, four of cecal origin and one from chicken meat, were not subjected to the antimicrobial resistance studies.

Phenotypic resistance pattern of *Campylobacter* isolates

A total of 103 isolates comprising 47 *C. jejuni* and 56 *C. coli* isolates were further studied. The species-wise and source-wise antimicrobial resistance patterns observed in the study are depicted in Tables 2 and 3. The isolates showed higher resistance against TET (64.1%), DOX (54.4%), AMP (46.3%), NA (42.7%), KAN (35.9%) and CIP (33.33%), and least resistance to COA (19.4%) and ERY (21.4%). Resistance to TET (68.1%), AMP (59.6%) and DOX (46.8%) was common among *C. jejuni* isolates, while *C. coli* isolates were resistant to TET and DOX (60.7% each), NA (57.14%) and CIP (35.7%). Both *C. jejuni* and *C. coli* showed lower levels of resistance against ERY (27.7% and 16.1%, respectively) and COA (19.1% and 19.6%, respectively). The chicken meat isolates were commonly resistant against TET (68.4%) and DOX (63.2%). A similar pattern was observed in isolates of cecal origin with maximum resistance to TET (61.4%) followed by DOX (48.6%). Isolates of chicken meat and cecal origin showed low resistance against ERY (26.3% and 24.3%, respectively) and moderate resistance against CIP (31.6% and 32.9%, respectively). The slaughter house swab and sewage isolates were highly resistant to AMP (78.6%), DOX (71.4%) and TET (71.4%), while all the isolates were susceptible to ERY.

Multi-drug resistance was observed in 54.37% ($n = 56$) of the isolates, with 53.2% (25/47) *C. jejuni* and 55.3% (31/56) *C. coli* isolates resistant against three or more classes of antimicrobials. Amongst the 35 ciprofloxacin resistant and 22 erythromycin resistant isolates, 60% (21/35) and 81.8% (18/22) isolates, respectively were MDR. Source wise analysis revealed that 68.4% (13/19) chicken meat isolates, 54.5% (6/11) slaughter house environment isolates, 48.57% (34/70) cecal origin isolates and all three sewage origin isolates were multi-drug resistant. The MAR index was in the range of 0.11–0.78, with maximum number of isolates resistant against 2, 3 or 4 antimicrobials. The MIC_{50} and MIC_{90} values for *C. jejuni and C. coli* isolates are listed in Table 2. The resistance profiles of the 56 MDR isolates are depicted in Table 4.

Molecular mechanism of antimicrobial resistance

The analysis of *gyr*A gene sequences of 22 fluoroquinolone resistant isolates (9 *C. jejuni* and 13 *C. coli*) revealed nucleotide substitution at codon 86 from ACA→ATA in *C. jejuni* and ACT→ATT in *C. coli*, which changed the amino acid from Threonine to Isoleucine (Thr-86-Ile). Silent mutations were observed in the resistant isolates at codons 82 (C→T), 120 (C→T), 127 (C→T) and 161 (T→C). The isolates

Table 4. Antimicrobial resistance patterns of multi-drug resistant *Campylobacter* spp. from broiler chicken and slaughter house environment in India.

			No. of isolates				
Sr. no.	No. of antibiotics/Class of antibiotic	Resistance pattern	Caec.[a] *Cj/Cc*[e]	CM[b] *Cj/Cc*	Sew.[c] *Cj/Cc*	SHE[d] *Cj/Cc*	Total *Cj/Cc*
1.	3/3	AMP-NA-ERY	0/1	0/0	0/0	0/0	0/1
2.	3/3	AMP-TET-ERY	1/0	0/0	0/0	0/0	1/0
3.	3/3	NA-TET- KAN	1/1	0/0	0/0	0/0	1/1
4.	3/3	CIP-ERY-KAN	1/1	0/0	0/0	0/0	1/1
5.	3/3	TET-ERY-KAN	2/0	1/0	0/0	0/0	3/0
6.	3/3	COA-NA-KAN	0/0	1/0	0/0	0/0	1/0
7.	3/3	AMP-CIP-DOX	0/1	1/0	0/0	0/0	1/1
8.	4/3	AMP-CIP-NA-KAN	0/0	0/1	0/0	0/0	0/1
9.	4/3	AMP-CIP-NA-TET	0/0	1/0	0/0	0/0	1/0
10.	4/3	NA-DOX-TET-KAN	0/3	0/0	0/0	0/1	0/4
11.	4/3	AMP-DOX-TET-KAN	1/2	0/1	0/0	0/0	1/3
12.	4/3	AMP-NA-DOX-TET	1/2	1/0	0/1	0/0	2/3
13.	4/3	CIP-DOX-TET-KAN	1/0	0/0	0/0	0/0	1/0
14.	4/3	COA-NA-DOX-TET	0/1	0/0	0/0	0/0	0/1
15.	4/3	AMP-COA- NA-DOX	0/1	0/0	1/0	0/0	1/1
16.	4/3	DOX-TET-ERY-KAN	2/0	0/0	0/0	0/0	2/0
17.	4/3	NA-DOX-TET-ERY	1/1	0/0	0/0	0/0	1/1
18.	5/3	COA-CIP-NA-DOX-TET	1/0	1/0	0/0	0/0	2/0
19.	5/3	AMP-COA-CIP-DOX-TET	0/0	0/0	0/1	0/0	0/1
20.	5/3	CIP-NA-DOX-TET-KAN	0/2	0/0	0/0	0/1	0/3
21.	5/3	AMP-CIP-NA-DOX-TET	0/1	0/1	0/0	0/0	0/2
22.	5/3	CIP-NA-DOX-TET-ERY	0/0	0/1	0/0	0/0	0/1
23.	5/4	AMP-NA-DOX-TET-KAN	0/0	0/0	1/0	0/0	1/0
24.	5/4	COA-DOX-TET-ERY-KAN	0/1	0/0	0/0	0/0	0/1
25.	5/4	COA-NA-DOX-TET-KAN	1/0	0/0	0/0	0/1	1/1
26.	6/3	AMP-COA-CIP-NA-DOX-TET	0/0	0/0	0/1	0/0	0/1
27.	6/4	AMP-COA-CIP-DOX-TET-ERY	1/0	0/0	0/0	0/0	1/0
28.	6/4	AMP-COA-CIP-DOX-TET-KAN	0/0	0/0	0/1	0/0	0/1
29.	6/4	AMP-COA- NA-DOX-TET-KAN	0/0	1/0	0/0	0/0	1/0
30.	6/5	COA-CIP-DOX-TET-ERY-KAN	1/0	0/0	0/0	0/0	1/0
31.	6/5	AMP-CIP-NA-TET-ERY-KAN	1/0	0/0	0/0	0/0	1/0
32.	7/5	AMP-COA-NA-DOX-TET-ERY-KAN	0/0	0/2	0/0	0/0	0/2
Species-wise total (*Cj/Cc*)[e]			16/18	7/6	2/4	0/3	25/31
Total			34	13	06	03	56

[a]Caec.: Poultry cecum; [b]CM: Chicken Meat; [c] Sew.: Sewage; [d]SHE: Slaughter house environment; [e]*Cj*: *C. jejuni*; [e]*Cc*: *C. coli*.

showing amino acid substitution had MIC values of 8 µg/ml and above for ciprofloxacin, and 32 µg/ml and above for nalidixic acid. None of the susceptible isolates showed Thr-86-Ile mutations. The macrolide resistance mechanism at the genotypic level was detected by identifying a nucleotide substitution (A→G) at the positions 2074 or 2075. The sequence of 23S rRNA of 19 ERY resistant isolates was analyzed to reveal mutations at A2074G in 12 isolates and at A2075G in one isolate. The 13 isolates with A to G mutation had an MIC of ≥64 µg/ml. No mutations were detected in any of the ERY susceptible isolates. The *tet*O gene was found in 64.1% (66/103) of the isolates. Among the DOX and/or TET resistant isolates, *tet*O gene was present in 91.7% (66/72) isolates, while none of the susceptible isolates harbored the gene. Among the *C. jejuni* and *C. coli* isolates, 91.2% (31/34) and 92.1% (35/38) possessed the *tet*O gene, respectively. The bla_{OXA-61} gene was present in 59.22% (61/103) isolates. The gene was found in 97.9% (47/48) of the AMP-resistant isolates. However, 14 isolates susceptible to AMP were also found to possess the gene. To the best of our knowledge, this is the first report from India on the presence of bla_{OXA-61} gene in *Campylobacter* spp. isolates.

Discussion

In the present study, 848 samples were screened for the presence of *Campylobacter* spp. and the overall occurrence of *Campylobacter* spp. was found to be 12.74%. The occurrence was highest among chicken cecum, followed by chicken meat and slaughter house environment swab samples. Presence of *C. jejuni* and *C. coli* at retail market meat samples poses risk to consumers. In similar studies on chicken meat sold at local retail shops, 36% of samples in Bareilly, India[8] and 29% in Lahore, Pakistan[29] were found to harbor *Campylobacter* spp. A higher prevalence of 56.1% has been reported in cecal samples from broilers at slaughter in China.[19] Other studies in India have reported isolation rates of 32%[30] and 15.89%[31] from chicken cecal samples. The isolation rate from slaughter house environment and sewage samples in this study was 10%. A study from China reported high contamination levels of 80–100% in swab samples from the operating table and 20–60% in workers' glove swabs.[32] Contamination rates of 89%[33] and 61%[34] have also been reported from slaughter houses. Presence of *Campylobacter* isolates on weighing scale, chopping boards and knives is a cause of concern since these sources can increase the chances of cross-contamination between carcasses. Retail shops in the unorganized sector usually carry out slaughter and processing of poultry on a single platform leading to cross-contamination between the carcasses and an increased risk of community transfer of campylobacteriosis.

The *Campylobacter* spp. isolates in this study showed higher resistance against tetracycline, doxycycline, ampicillin, nalidixic acid, kanamycin and ciprofloxacin, and low level of resistance to co-amoxiclav and erythromycin. The *C. jejuni* and *C. coli* isolates showed a similar trend of high resistance against tetracycline, and less common resistance against erythromycin and co-amoxiclav. Previous studies have reported that *C. coli* isolates possess a faster resistance gaining ability against a number of antimicrobial agents in comparison to *C. jejuni*.[35] However, no significant difference was observed in resistance levels of *C. jejuni* and *C. coli* in this study. A similar trend showing higher level of resistance against tetracycline (59.4%) and low level against erythromycin (6.9%) has been reported in isolates from India.[8] A higher resistance to tetracycline (73.5%), followed by ciprofloxacin (53%) and nalidixic acid (47.1%) was reported in *Campylobacter* species isolated from avian eggs in Iran.[36] The same study reported low level of resistance to erythromycin (5.9%). Resistance against tetracycline in isolates of poultry origin has been commonly reported.[18,19,32] A low level of resistance against erythromycin (17.59%) and moderate resistance against ciprofloxacin (33.33%) was recorded in our study. In contrast, many studies have reported high resistance ranging from 76.47% to 100% against ciprofloxacin[18,37,38] and 44.8% against erythromycin[39] in poultry isolates. Han et al.[19] reported erythromycin resistance in 93.9% *C. coli* isolates and 14% *C. jejuni* isolates. The slaughter house swab and sewage isolates were highly resistant to ampicillin, doxycycline, tetracycline and kanamycin. The antimicrobial usage pattern for therapeutic and non-therapeutic purposes differs by regions and the same may be responsible for the variation observed in resistance. Nevertheless, any level resistance against erythromycin and ciprofloxacin in *Campylobacter* spp. must be viewed with caution.

Multi-drug resistance was observed in 54.37% of the isolates screened in this study, with 53.2% of *C. jejuni* and 55.3% of *C. coli* isolates resistant against three or more class of antimicrobials. Occurrence of higher percent of MDR strains (75.9–93.7%) has been reported from poultry sources.[3,19,30] Studies in India have reported 31.6% MDR strains in poultry isolates[8] and 30.6% in human isolates.[40] Lehtopolku[41] reported that 49% *C. jejuni* and 54% *C. coli* isolates from Finland were multi-drug resistant during the period 2003–2005 with an overall MDR occurrence of 16%. Source wise analysis revealed that 68.4% (13/19) chicken meat isolates, 64.3% (9/14) slaughter house environment and sewage isolates, and 48.6% (34/70) cecal origin isolates were multi-drug resistant. Amongst the 35 ciprofloxacin resistant and 22 erythromycin resistant isolates, Amongst the 35 ciprofloxacin resistant and 22 erythromycin resistant isolates, 60% (21/35) and 81.8% (18/22) isolates, respectively were MDR. The increasing occurrence of multidrug resistant *Campylobacter* strains may be attributed to overuse of antimicrobial agents as therapeutics and growth promoters in animals and poultry.[42] The occurrence of relatively high level of multidrug resistance to most of the antibiotics examined is a concern as it might elicit the possibility of clinical treatment failures in cases of severe *Campylobacter* infection. Limited information exists on the treatment of infections caused by MDR strains of *Campylobacter* spp. In this scenario, it is apt to aim at rationalizing the use of antimicrobials in livestock and poultry.

Amino acid substitution at codon 86 (Thr-86-Ile) was observed in fluoroquinolone resistant strains along with silent mutations at codons 82 (C→T), 120 (C→T), 127 (C→T) and 161(T→C). The isolates showing amino acid substitutions had MIC values of $\geq 8\,\mu g/ml$ for ciprofloxacin, and $\geq 32\,\mu g/ml$ for nalidixic acid. Mutations in *gyrA* have been reported in ciprofloxacin and nalidixic acid resistant isolates.[18,19] Chatur et al.[43] studied fluoroquinolone resistant isolates from India and reported point mutations at codon 86 in all the resistant isolates with MICs of 4–16 μg/ml for ciprofloxacin and 16–128 μg/ml for nalidixic acid. The same study reported silent mutations at codon 81 (C→T), 119 (C→T) and 121 (T→C). Similar observations of Thr-86-Ile point mutation were made by Alonso et al.[44] who also observed MICs from 16 to 64 μg/ml. Resistance to fluoroquinolones in *Campylobacter* is mainly attributed to mutation in the QRDR of *gyrA*.[45] The Thr-86-Ile substitution was the most frequently observed amino acid change and has also been reported as a common occurrence previously.[46–48]

A total of 19 erythromycin resistant isolates were subjected to mutation analysis, which revealed that 12 isolates had a nucleotide substitution at position 2074 and 1 isolate at 2075. All the above 13 isolates showed higher MIC values of $\geq$64 µg/ml against erythromycin. No mutation was detected in any of the susceptible isolates. An earlier study on poultry origin isolates reported A2075G mutation in 11 resistant strains with MICs ranging from 64 to $\geq$512 µg/mL.[49] A high number (96.4%) of erythromycin-resistant isolates were found to have a mutation at A2075G.[20] The transitional mutation at A2075G is commonly reported to be associated with high-level resistance to erythromycin among *C. coli* and *C. jejuni* isolates.[50–52] Further, presence of *erm*(B) may also have contributed to the resistance against erythromycin.

The presence of *tetO* gene was confirmed in 91.7% (68/72) of the doxycycline and tetracycline resistant isolates while it was absent in all the susceptible isolates. However, Hungaro et al.[53] and Obeng et al.[54] reported low correlation between resistance and presence of the gene, with *tetO* gene present in only 40% of the tetracycline resistant isolates. On the contrary, some studies have reported presence of *tetO* gene in all tetracycline resistant isolates.[45,55] The presence of *tet*O gene in ~92% of the tetracycline resistant isolates reaffirms the fact that tetracycline resistance in *Campylobacter* spp. is predominantly associated with this gene.

The $bla_{\text{OXA-61}}$ gene was detected in 59.22% (61/103) of the isolates, with 97.9% (47/48) ampicillin resistant isolates possessing the gene. Griggs et al.[28] made a similar observation with 91% ampicillin-resistant isolates carrying $bla_{\text{OXA-61}}$ gene. Among the isolates from different sources, sewage origin isolates showed highest presence of $bla_{\text{OXA-61}}$ gene (81.82%) followed by chicken meat (80%). The $bla_{\text{OXA-61}}$ gene was also detected in 14 ampicillin susceptible isolates and similar findings were reported that the ampicillin susceptible isolates from chicken were able to encode the $bla_{\text{OXA-61}}$ gene.[53,54]

Conclusion

Presence of *Campylobacter* spp. isolates in chicken meat samples and slaughter house environment is a cause of concern since these sources can increase the chances of cross-contamination between carcasses. Retail shops usually carry out slaughter and processing of poultry on a single platform leading to cross-contamination between the carcasses and hence an increased risk of community transfer of campylobacteriosis. The results in the present study revealed high degree of resistance to tetracycline antimicrobials, followed by penicillin and fluoroquinolones. The erythromycin resistance was at a lower level. The rise in MDR strains of *Camylobacter* spp. is a major cause of concern. The various molecular mechanisms of antimicrobial resistance in *Campylobacter* spp. were also studied. The occurrence of antimicrobial resistant *Campylobacter* spp. calls for judicial use of antimicrobials in animal production. Further, the use of alternatives to antimicrobials such as probiotics and bacteriophages as a means of minimizing bacterial load in the poultry intestine should be evaluated and applied at the farm level.

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